# JOURNEY RISK MANAGEMENT (JRM) STUDY

## Salem Terminal TO MANGAI FUELS

## Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

## Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

## Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

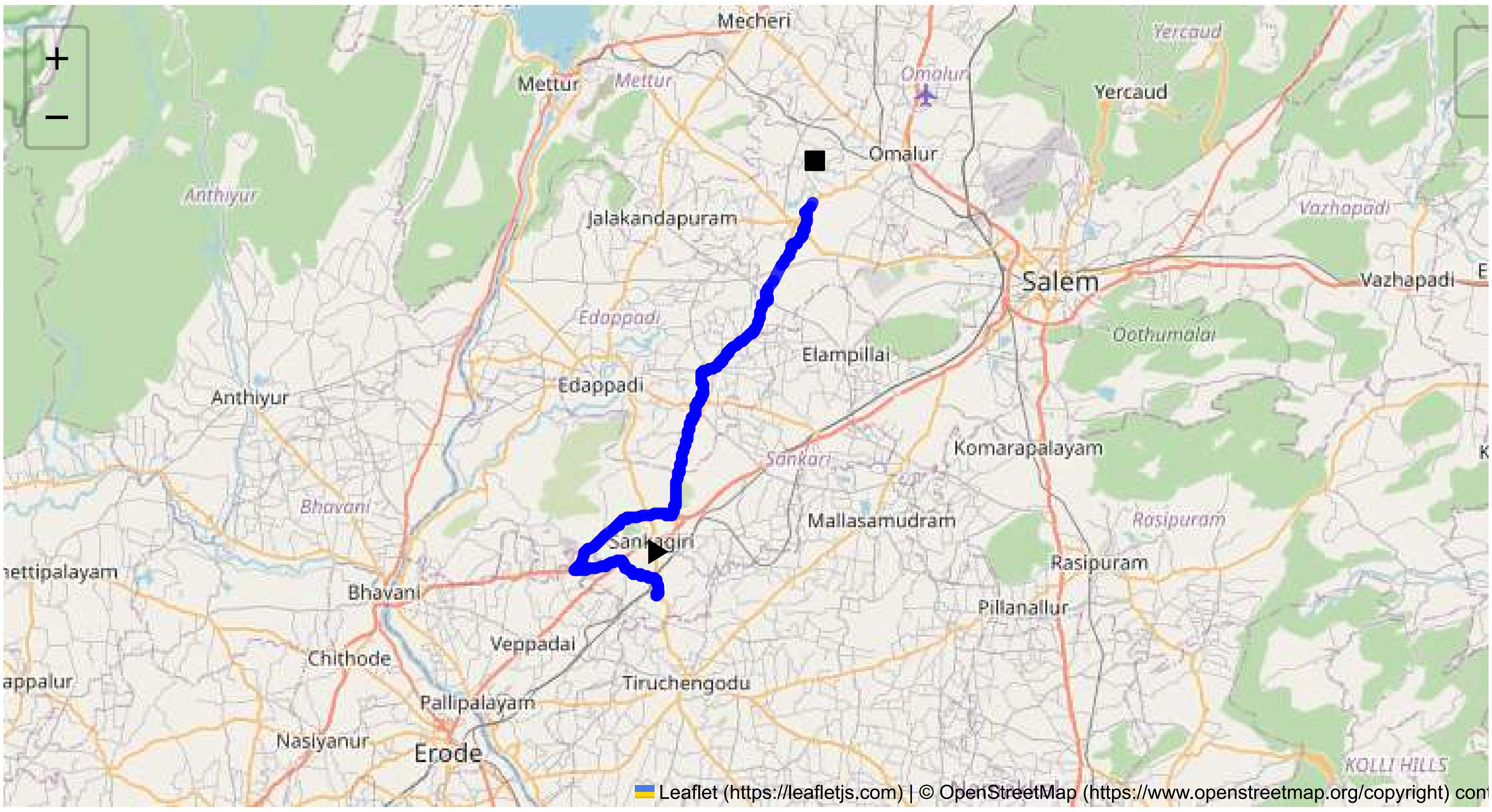
## Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

## Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.


Route Summary:
Total Distance: 45.56 km
Estimated Duration: 1.1 hours
Adjusted Duration (Heavy Vehicle): 1.4 hours
Start: (11.4381, 77.8734)
End: (11.70794, 77.984156)




## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

The route from Sangagiri to Sikkampatti spans approximately 45.56 kilometers and generally travels through rural and semi-urban regions of Tamil Nadu. Starting from Sangagiri, it passes through key transition points such as Puthur, Padaiveedu, Erumaipatti, and Desavilakku before reaching Sikkampatti. The journey involves traversing local roads, state highways, and potentially busy junctions.

## 2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate with distinct wet and dry seasons. The area can experience heavy rainfall during the monsoon season (June to September), which may lead to localized flooding and reduced visibility. Summers can be extremely hot, potentially causing road surface damage or heat-related vehicle malfunctions. Fog is also possible in the winter months, affecting visibility.

## 3. Analysis of Traffic Patterns

Traffic patterns vary across the route:

- **Morning and Evening Rush Hours**: Congestion tends to peak during morning (8-10 AM) and evening (5-7 PM) times, particularly in and around urbanized zones like Salem Highway.
- **Congestion-Prone Areas**: Potential bottlenecks can be expected near larger towns such as Padaiveedu and segments along the Salem - Kochi Highway due to commuter traffic and market areas.

## 4. Assessment of Road Quality and Infrastructure

The road conditions vary:

- **Salem - Kochi Highway**: Well-maintained, but narrow lanes and sharp turns in some areas may pose challenges.
- **Local and Village Roads**: These may have uneven surfaces, potholes, and lack proper signage or lighting.

## 5. Suggestions for Alternative Routes for Emergencies

In case of blockages or emergencies:

- **Primary Route**: Use alternate routes towards the NH544 near Salem that heads northwards and bypasses heavily congested areas.
- **Local Detours**: Consider village roads that run parallel to major highways, although these might increase travel time and feature poor road conditions.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

Transporting hazardous materials requires adherence to the Motor Vehicles Act, 1988, with specific regulations on placarding, vehicle markings, and driver qualifications. Night movement of hazardous material may have additional restrictions in some areas.

## 7. Overview of Historical Incidents

There have been periodic reports of heavy vehicle accidents along this stretch due to speeding, hazardous overtaking on narrow roads, and during night operations. No significant incidents specifically involving hazardous materials have been documented, but caution is advised due to general road conditions.

## 8. Environmental Considerations and Sensitive Areas

The route crosses certain ecologically sensitive areas, including agricultural land, requiring care to avoid spills that could harm local ecosystems. Impact on water bodies during the rainy season is a significant concern.

## 9. Analysis of Communication Coverage

The area generally has good communication coverage with few potential dead zones, particularly in more rural stretches. However, service interruptions are possible due to network congestion or infrastructure damage during adverse weather.

## 10. Estimated Emergency Response Times

Emergency services, such as ambulances and fire brigades, are primarily located in larger towns. Estimated response times:

- **Urban Centers**: 15-30 minutes
- **Rural or Remote Areas**: 30-60 minutes, potentially longer depending on the severity of incidents and road conditions.

## 11. Overall Summary of Risk Assessment

The route presents a moderate level of risk for transporting hazardous materials. Key risks include weather-induced hazards, variable road quality, and traffic congestion. Communicating these risks effectively to drivers and ensuring compliance with local regulations should minimize incidents. Coordination with local emergency services and having a real-time communication plan are recommended to improve safety measures. Continuous monitoring of road conditions and updated traffic information will aid in adaptive route management.

## Risk Assessment - Turns

| | Risk Type | Risk Level | Coordinates | Speed Limit |
|---|---|---|---|---|
| 0 | Turn | High | 11.43811, 77.87348 | 15 KM/Hr |
| 1 | Turn | Medium | 11.43961, 77.87341 | 30 KM/Hr |
| 2 | Turn | Medium | 11.43968, 77.87345 | 30 KM/Hr |
| 3 | Turn | High | 11.44029, 77.87544 | 15 KM/Hr |
| 4 | Turn | Medium | 11.44898, 77.87410 | 30 KM/Hr |
| 5 | Turn | Medium | 11.45357, 77.85789 | 30 KM/Hr |

| | Risk Type | Risk Level | Coordinates | Speed Limit |
|---|---|---|---|---|
| 6 | Turn | Medium | 11.45352, 77.85698 | 30 KM/Hr |
| 7 | Turn | Medium | 11.46303, 77.84945 | 30 KM/Hr |
| 8 | Turn | Medium | 11.46318, 77.84913 | 30 KM/Hr |
| 9 | Turn | High | 11.45542, 77.81725 | 15 KM/Hr |
| 10 | Turn | High | 11.45631, 77.81668 | 15 KM/Hr |
| 11 | Turn | High | 11.49409, 77.88004 | 15 KM/Hr |
| 12 | Turn | Medium | 11.49419, 77.88015 | 30 KM/Hr |
| 13 | Turn | High | 11.49383, 77.88442 | 15 KM/Hr |
| 14 | Turn | Medium | 11.49419, 77.88454 | 30 KM/Hr |
| 15 | Turn | High | 11.49448, 77.88490 | 15 KM/Hr |
| 16 | Turn | Medium | 11.56083, 77.89921 | 30 KM/Hr |
| 17 | Turn | Medium | 11.57048, 77.90224 | 30 KM/Hr |
| 18 | Turn | Medium | 11.58304, 77.90656 | 30 KM/Hr |
| 19 | Turn | Medium | 11.63840, 77.94820 | 30 KM/Hr |
| 20 | Turn | High | 11.67868, 77.96836 | 15 KM/Hr |
| 21 | Turn | Medium | 11.68913, 77.97734 | 30 KM/Hr |
| 22 | Turn | Medium | 11.68925, 77.97741 | 30 KM/Hr |
| 23 | Turn | High | 11.68927, 77.97747 | 15 KM/Hr |
| 24 | Turn | High | 11.70223, 77.97860 | 15 KM/Hr |

## Emergency Locations

| | type | name | coordinates | speed_limit | risk_level |
|---|---|---|---|---|---|
| 1 | hospital | Sri Shanthi Hospital | 11.569272, 77.901937 | 30 km/h | Medium |

## Crowded Spots

| | type | name | coordinates | speed_limit | risk_level |
|---|---|---|---|---|---|
| 0 | school | KRP Matric. Hr. Sec School | 11.4546193, 77.8142445 | 30 km/h | Medium |

# Route Photos of Risky Spots


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.44029, 77.87544


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.44898, 77.87410


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45357, 77.85789


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.45352, 77.85698

**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46303, 77.84945


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.46318, 77.84913


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45542, 77.81725


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.45631, 77.81668


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49409, 77.88004

**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49419, 77.88015

**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49383, 77.88442

**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.49419, 77.88454


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.49448, 77.88490


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.56083, 77.89921

**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.57048, 77.90224


**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.58304, 77.90656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**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.68913, 77.97734

**Risk Type:** Turn

**Risk Level:** Medium

**Speed Limit:** 30 KM/Hr

**Coordinates:** 11.68925, 77.97741

**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.68927, 77.97747


**Risk Type:** Turn

**Risk Level:** High

**Speed Limit:** 15 KM/Hr

**Coordinates:** 11.70223, 77.97860